

Time limit: 20 minutes, strict. Start a timer. Stop when it rings, finished or not. Answers are on page 2: use them if you are stuck, and say so in the email. The point is to try.

What to send, by Tuesday 6 October, 20:00: one email to Gauti.Eggertsson@gmail.com, subject “Sprint 1”, three lines:

Done at [time]. Got to problem [n]. Looked at answers for [which, or none].

On the weekend you redo these with Gauti over Zoom, without the answers.

1. Solve $x^2 - 3x - 4 = 0$. Then solve $x^2 - 3x - 4 > 0$ and $x^2 - 3x - 4 \leq 0$. Read the inequalities off a sketch.

2. $f(x) = 2x^2 - 12x + 18$ and $g(x) = 2x^2 - 12x + 23$. For each: the discriminant, the roots (if any), the vertex, and the vertex form $a(x - h)^2 + k$.

3. A parabola $y = f(x)$ has x -intercepts 2 and 6, y -intercept 12, and passes through $A(7, 5)$. (a) Write down the axis of symmetry. (b) Find $f(8)$ and $f(1)$ using symmetry only. (c) Find $f(x)$ in the form $ax^2 + bx + c$.

4. $f(x) = 2x^2 - 4x + k$. (a) For which k does $f(x) = 0$ have exactly one root? (b) For which k is $f(x) > 0$ for all x ? (c) For which k is $f(x) \geq 0$ for all x ?

Look only when stuck, and say which ones in the email. On the weekend these are redone without this page.

1. $(x - 4)(x + 1) = 0$: $x = -1$ or $x = 4$. Opens upward, so > 0 outside the roots: $x < -1$ or $x > 4$. ≤ 0 between them: $-1 \leq x \leq 4$.
2. f : $\Delta = 144 - 144 = 0$, one repeated root $x = 3$, vertex $(3, 0)$, $f(x) = 2(x - 3)^2$. g : $\Delta = 144 - 184 = -40 < 0$, no real roots, vertex $(3, 5)$, $g(x) = 2(x - 3)^2 + 5$, always positive.
3. (a) Halfway between the roots: $x = 4$. (b) 8 mirrors 0, so $f(8) = f(0) = 12$; 1 mirrors 7, so $f(1) = f(7) = 5$. (c) $f(x) = a(x - 2)(x - 6)$ with $f(0) = 12a = 12$, so $a = 1$: $f(x) = x^2 - 8x + 12$. Check $f(7) = 49 - 56 + 12 = 5$.
4. $\Delta = 16 - 8k$. (a) $\Delta = 0$: $k = 2$. (b) Opens upward ($a = 2 > 0$) and $\Delta < 0$: $k > 2$. (c) $\Delta \leq 0$: $k \geq 2$. The small line under the sign is the whole difference.